1. Data was collected about various students. Part of this data can be seen in the table.

1 point

What types of variables are Gender and Age?

Gender	Age (in years)
F	18
M	23
M	29
M	31
F	39

- Gender is binary and Age is categorical.
- Gender is categorical and Age is numerical.
- Gender and Age are both categorical.
- Gender and Age are both numerical.

2. We have measured two variables: the colour and the size of a product, as seen in the table.

1 point

What types of variables are these?

Colour	Size
Green	Small
Blue	Medium
Blue	Large
Red	Large
Blue	Medium

- Colour is ordinal and Size is ordinal.
- Colour is nominal and Size is nominal.
- Colour is oridnal and Size is nominal.
- Colour is nominal and Size is ordinal.

3. You collected data on the time it took call center employees to handle a phone call. The gathered data is displayed in the table.

1 point

Which of these two variables contains more information?

Total handling time (seconds)	Done within 5 mins?
332	No
778	No
361	No
794	No
231	Yes
375	No
841	No
596	No
124	Yes
224	Yes

- Done within 5 minutes, because it is numerical.
- Total handling time, because it is categorical.
- Total handling time, because it is numerical.
- Done within 5 minutes, because it is categorical.

1 point

Descriptive Statistics: Work hours

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Work hours	30	1	6.733	0.224	1.225	4.600	5.475	6.900	8.125	9.400

4. You have measured the number of 'work hours' for various employees on a certain day. See the displayed descriptive statistics.

The norm for work hours is 8 hours per day, that is, all employees are required to work at least 8 hours per day.

Which statement is **false**?

- On average employees do not meet the norm of 8 hours per day.
- One employee worked one hour (N* = 1).
- At least 25% of the employees worked more than the norm of 8 hours.
- All employees worked at least 4.6 hours.

1 point

Descriptive Statistics: Work hours

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Work hours	30	1	6.733	0.224	1.225	4.600	5.475	6.900	8.125	9.400

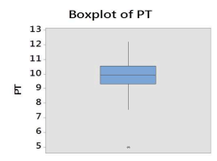
5. You have measured the number of work hours for various employees on a certain day. You computed the displayed descriptive statistics.

The norm for work hours is 8 hours per day, that is, all employees are required to work at least 8 hours per day.

Which statement is **false**?

- The dispersion equals 0.224.
- 50% of the workdays lasted more than 6.9 hours.
- 25% of the workdays was shorter than 5.475 hours.
- The range is equal to 4.8 hours.

1 point

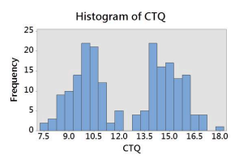


6. The norm for the processing times equals 10 minutes. That is, the processing time should be less than or equal to 10 minutes.

What is **false** about the processing times based on the boxplot?

- The middle line in the box shows the average.
- There is an outlier in the data.
- Around 50% of the data is below the norm.
- The lowest measured value is around 5.

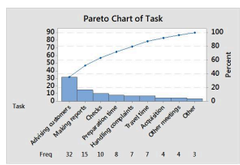
1 point



7. You have measured your CTQ. What can be concluded based on this histogram?

- The data is a mixture of distributions.
- The data contain many outliers.
- The data is symmetrical.
- The data is skewed.

1 point



8. As part of an improvement project, you want to establish what type of activities employees undertake.

For this, the team counted the tasks they performed. The results are displayed in the Pareto chart.

What can be concluded based on this chart?

- Advising customers takes up most of the employees' time.
- Two activities are approximately 50% of all performed tasks.
- In total 100 tasks were measured.
- Nothing yet, since the category "Other" has to be specified first.

1 point

9.

Production Volume	Speed
85	3
90	4
120	3
150	5

You have measured the variables Production Volume and Machine Speed. Part of the data is displayed in the table.

You are interested to display both variables in one figure. Which figure is most suitable to visualize these two variables together?

- Two sample t-test.
- Scatterplot.
- Pareto chart.
- Multiple boxplot.

1 point

10. Revenue per week was measured for two shops in different locations.

What **cannot** be concluded based on the multiple boxplot that displays the collected data?

- Location A had one week with exceptional revenue.
- More data was collected for Location B, as the box is considerably larger.
- The median revenue of location B is higher than the median revenue of location A.
- There seems to be a difference in dispersion beween the two locations.